From Person-Specific Dynamics to Population Inference: A Computationally Efficient Meta-Analytic Structural Equation Modeling Framework for Integrating Complex Multilevel Dynamic Models

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This research was supported by the Prevention and Methodology Training Program (PAMT), through a T32 training grant from the National Institute on Drug Abuse (NIDA; T32 DA017629; MPIs: J. Maggs and S. Lanza), as well as by grant UL1TR002014-06 from the National Center for Advancing Translational Sciences (NCATS).

Computations for this research were performed on the Pennsylvania State University's Institute for Computational and Data Sciences' Roar supercomputer using SLURM for job scheduling (Yoo et al., 2003), GNU Parallel to run the simulations in parallel (Tange, 2021), and Apptainer to ensure a reproducible software stack (Kurtzer et al., 2017, 2021).

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Abstract

Intensive longitudinal research often seeks population-level inference on within-person dynamics without sacrificing idiographic estimation. We introduce MetaVAR, a two-stage meta-analytic structural equation modeling (MASEM) framework for multilevel dynamic systems that combines person-specific dynamic modeling with multivariate meta-analytic synthesis. In Stage 1, person-specific vector autoregressive (VAR) models are fit to obtain each individual's dynamic parameter vector and corresponding sampling covariance matrix. In Stage 2, those parameter vectors are synthesized using a multivariate random-effects meta-analytic structural equation model, allowing population-level inference on average dynamics, between-person heterogeneity, and cross-parameter covariation while propagating first-stage uncertainty. A 36-condition simulation compares MetaVAR with Bayesian dynamic structural equation modeling (DSEM) under default and alternative priors and an uncertainty-uncorrected two-stage benchmark, varying sample size, series length, and transition heterogeneity, including zero heterogeneity and unequal lengths. MetaVAR's clearest advantages concerned between-person transition heterogeneity: uncertainty propagation substantially improved recovery over the uncorrected approach, and transition-variance coverage exceeded both DSEM branches, particularly at smaller nonzero heterogeneity. DSEM generally recovered population-average transition effects better. A dedicated benchmark found MetaVAR approximately 9.3 times faster than DSEM under the evaluated configuration. An ecological momentary assessment illustration shows interpretable average affect dynamics and heterogeneity across a nine-feature system. MetaVAR provides an efficient route from uncertain idiographic estimates to multivariate population inference.

Keywords: vector autoregressive model, multilevel model, random-effects meta-analysis, structural equation modeling

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Intensive longitudinal designs, such as ecological momentary assessment (EMA), experience-sampling, daily diary, ambulatory assessment, and measurement-burst protocols, have made it increasingly feasible to collect dense within-person data in everyday contexts (Bolger & Laurenceau, 2013; Bolger et al., 2003; Csikszentmihalyi & Larson, 1987; Nesselroade, 1991; Shiffman et al., 2008; Sliwinski, 2008; Trull & Ebner-Priemer, 2013). These designs have expanded opportunities to study psychological processes as dynamic systems, in which trajectories evolve over time within persons and individuals differ meaningfully in the parameters governing those trajectories (Asparouhov et al., 2018; Bringmann et al., 2013, 2016; Fisher et al., 2011; Hamaker et al., 2018; Rowland & Wenzel, 2020). This shift reinforces a long-standing methodological point: inferences about within-person dynamics cannot be assumed from between-person variation, and questions about regulation, coupling, and stability require models that distinguish idiographic (person-specific) from nomothetic (population-level) structure (Curran & Bauer, 2011; Fisher et al., 2018; Gates et al., 2023; Hamaker et al., 2005; Hunter et al., 2024; Molenaar, 2004; Wang et al., 2026).

At the same time, applied researchers rarely want only a collection of person-specific estimates. They also want population-level summaries of average dynamics, between-person heterogeneity, and the ways dynamic features co-vary across people, as well as a basis for relating those person-specific dynamic signatures to covariates and distal outcomes (Beltz et al., 2016; Li et al., 2022). Existing workflows, however, often force a trade-off between two unsatisfactory options. One-step hierarchical models with partial pooling—including multilevel and Bayesian multilevel formulations—can support population inference, but may become computationally demanding and shrink person-specific dynamics toward the population mean. At the other extreme, fully unpooled "fit-many-then-summarize" workflows are modular and scalable, but uncertainty-uncorrected versions often ignore first-stage uncertainty and the multivariate dependence among dynamic parameters, limiting valid inference on the dynamic system as a whole (Moeyaert et al., 2017; Van den Noortgate & Onghena, 2008).

This paper addresses this gap by introducing MetaVAR, a two-stage meta-analytic structural equation modeling framework (MASEM; Cheung, 2008, 2014; Cheung & Cheung, 2016) for multilevel dynamic models. In this manuscript, the Stage 1 idiographic model is a vector autoregressive model (VAR; Hamilton, 1994; Lütkepohl, 2005) fit separately to each individual, and Stage 2 synthesizes the resulting person-specific parameter vectors using a multivariate meta-analytic structural equation model (SEM) with random effects. The key methodological contribution is not simply that MetaVAR is two-stage, but that it

synthesizes the full person-specific dynamic parameter vector while carrying forward the full Stage 1 sampling covariance matrix. This enables population-level inference on average dynamics, between-person heterogeneity, and cross-parameter covariation in the joint dynamic system rather than on isolated parameters one at a time. A further advantage of this formulation is that Stage 2 supports system-level hypotheses on the full dynamic parameter vector—including equality constraints, structured relations among parameters, and links to covariates or distal outcomes—rather than limiting inference to isolated coefficients considered one at a time.

The central question, then, is not only how to pool person-specific dynamic estimates, but how to do so in a way that preserves inferential validity while remaining practically feasible. Existing uncertainty-uncorrected two-stage workflows can compromise this validity by ignoring first-stage estimation error and the multivariate dependence among person-specific dynamic parameters, whereas one-step hierarchical models can be computationally demanding and partially pool person-specific estimates toward the population mean. MetaVAR is intended for settings in which researchers want to retain idiographic first-stage estimation while still obtaining valid multivariate population-level inference. In particular, MetaVAR is most useful when the scientific target includes not only population-average dynamics, but also between-person heterogeneity and cross-parameter covariation in the dynamic system. This practical niche is especially relevant in applications where person-specific models can be estimated independently and in parallel, making a two-stage workflow computationally attractive relative to joint hierarchical estimation. In what follows, we situate this contribution relative to existing one-step and two-stage approaches, evaluate it in a Monte Carlo simulation against a Bayesian multilevel VAR benchmark, operationalized here using dynamic structural equation modeling under default and alternative priors in `Mplus`[1], alongside an uncertainty-uncorrected two-stage fit-many-then-summarize approach, and illustrate it using ecological momentary assessment data on negative and positive affect.

Two Dominant Strategies and a Persistent Gap

Two broad strategies dominate current practice for drawing population-level conclusions from intensive longitudinal multivariate time series. The first is hierarchical one-step (partially pooled) modeling—including multilevel and Bayesian multilevel formulations such as dynamic structural equation modeling (DSEM) and Bayesian multilevel VAR—in which within-person dynamics and between-person variation are estimated simultaneously within a single hierarchical model (Asparouhov et al., 2018; Hamaker et al., 2018; Li et al., 2022). This approach is attractive in behavioral sciences because the length

[1] We treat this as a practical benchmark under two specific prior configurations in `Mplus` rather than as a definitive evaluation of all Bayesian multilevel VAR specifications. In one-step Bayesian estimation, performance can depend materially on prior specification for random-effects covariance structures (Asparouhov & Muthén, 2024b; Li et al., 2022; Schuurman et al., 2016; van Erp et al., 2018).

of the series per-person is often limited, making it useful to borrow strength across individuals while still allowing random effects in dynamic parameters (Li et al., 2022). At the same time, the estimation problem can become computationally demanding as the number of variables and random effects grows, and practical implementations often require simplifying assumptions about the random-effects structure or otherwise face feasibility and convergence constraints in higher-dimensional settings (Asparouhov et al., 2018; Epskamp et al., 2018; Li et al., 2022). Moreover, because hierarchical estimators partially pool information, person-specific dynamics are shrunk toward the population mean. Although this can improve stability and out-of-sample performance (Gelman & Hill, 2006), it also means that person-level estimates are biased toward the center and may be less suitable as direct measures of individual differences or for detecting unusual individuals (Efron & Morris, 1975).

At the other extreme are fully unpooled workflows that fit an idiographic model per person and then summarize the resulting coefficients informally, for example, by averaging. This "fit-many-then-summarize" strategy is modular and naturally parallelizable, but uncertainty-uncorrected summaries often treat person-specific estimates as if they were equally precise or measured without error, even though their precision can differ as a function of series length, missingness, and measurement quality (Hedges & Olkin, 1985; Pastor & Lazowski, 2018). In intensive time-series settings, dependence and model misspecification (e.g., unmodeled trends) can also affect precision and inference if not modeled (Baek et al., 2023; Shadish et al., 2008). A distinct limitation arises in multivariate dynamic systems: even when first-stage uncertainty is acknowledged, many multi-stage approaches do not retain the within-person sampling covariance matrix among dynamic parameters, making it difficult to synthesize the joint dynamic system at the second stage (Epskamp et al., 2018; Moeyaert et al., 2017). Valid two-stage inference therefore requires carrying forward each individual's first-stage sampling variance and, in multivariate settings, the full sampling covariance matrix rather than treating person-level estimates as error-free (Jackson et al., 2011; Riley, 2009).

Taken together, this basic trade-off between borrowing strength and idiographic flexibility is consistent with earlier comparisons of person-specific and multilevel autoregressive modeling, which showed that multilevel estimation can be advantageous when dynamic structure is relatively homogeneous or per-person series are short, whereas person-specific modeling becomes more attractive as dynamic heterogeneity increases and sufficient repeated observations are available (Liu, 2017). These trade-offs motivate a more principled two-stage alternative: estimate person-specific dynamic models first, then synthesize those results at the population level while explicitly accounting for first-stage uncertainty and dependence among estimated parameters. The gap is therefore not simply the absence of a two-stage

workflow, but the absence of a two-stage workflow that supports valid multivariate population inference on the dynamic system as a whole.

Related Literatures: Idiographic Two-Stage Dynamic Modeling and Meta-Analytic SEM

MetaVAR brings together ideas from two related bodies of literature. The first is a growing literature on two-stage idiographic modeling for intensive longitudinal data, especially in dynamic network and related multivariate time-series traditions. Lee and Gates (2024) advocated a two-stage idiographic meta-analytic perspective for observational time series, emphasizing the importance of estimating within-person models idiographically and then using random-effects meta-analysis to characterize population distributions of within-person effects. Similarly, Zhang et al. (2025) showed, in the context of individual-level dynamic factor analysis (DFA), that ignoring first-stage estimation error can bias second-stage random-effects estimation. Taken together, this literature establishes both the scientific value of treating heterogeneity as a target of inference and the statistical importance of carrying first-stage uncertainty into second-stage synthesis.

The second body of literature is meta-analytic structural equation modeling (Cheung, 2008, 2014, 2018; Cheung & Cheung, 2016; Jak & Cheung, 2020). MASEM is attractive here because it can fit full path or factor models, account for sampling covariance among effect sizes, provide overall model fit, model heterogeneity in multivariate form, and accommodate dependent effect sizes in multilevel formulations (Cheung, 2008, 2014; Cheung & Cheung, 2016). SEM-based meta-analysis has also been used to compute multivariate effect sizes together with their sampling covariance matrices and to impose, test, or relax equality constraints as needed (Cheung, 2018). These are precisely the features needed for a Stage 2 model whose unit of inference is the full dynamic parameter system rather than one coefficient at a time. MetaVAR integrates these two literatures by embedding idiographic dynamic-model outputs within a multivariate meta-analytic SEM framework (Mehta & Neale, 2005).

More broadly, this framing aligns with a longer tradition of synthesizing person-specific or single-case studies. Earlier work discussed meta-analysis of applied time-series data; later work in single-case research developed hierarchical and multilevel methods for combining effect sizes across cases; and more recent work has extended those approaches to incorporate time, moderators, multiple outcomes, and heterogeneous within-case variance (Baek et al., 2023; Gingerich, 1984; Moeyaert et al., 2020; Ugille et al., 2012; Van den Noortgate & Onghena, 2003). MetaVAR extends this general logic to person-specific multilevel dynamic models by treating each individual's estimated dynamic signature as a multivariate effect-size vector to be synthesized jointly.

The move from path-specific synthesis to multivariate synthesis is not merely a technical generalization. In dynamic systems, substantive questions often concern configurations of parameters

rather than isolated coefficients—for example, whether stronger autoregressive persistence co-occurs with stronger cross-lagged coupling, whether coupling is asymmetric across directions, or whether summary features of the system vary systematically across persons. Addressing such questions requires a second stage that treats the dynamic system itself as the unit of inference. MetaVAR is designed for that purpose.

MetaVAR: Dynamic MASEM for System-Level Pooling With Full Error Propagation

MetaVAR is a two-stage framework that treats multilevel dynamic modeling as a meta-analytic problem in which *persons play the role of studies*. In Stage 1, an idiographic dynamic model is fit separately to each individual $i = 1, \ldots, N$, outside the Stage 2 random-effects SEM, yielding a vector of estimated dynamic parameters and an accompanying sampling covariance matrix. In Stage 2, these person-level parameter vectors are synthesized using a multivariate random-effects meta-analytic model expressed as a structural equation model. The key feature of this design is that the full Stage 1 sampling covariance matrix is carried forward into Stage 2. As a result, MetaVAR supports population-level inference on the *joint* dynamic parameter system, including average dynamics, between-person heterogeneity, and cross-parameter covariation.

This perspective connects directly to multivariate random-effects meta-analysis (Berkey et al., 1998; van Houwelingen et al., 1993, 2002) and MASEM (Cheung, 2008, 2013, 2014; Cheung & Cheung, 2016). It is motivated by questions that arise frequently in intensive longitudinal research: What is the average within-person dynamic system? How much do people differ in that system? Which dynamic features tend to co-occur across people? More broadly, the same framework provides a basis for extensions involving person-level covariates or distal outcomes, although the present manuscript focuses on the core two-stage synthesis.[2]

Stage 1 Outputs as Multivariate Effect Sizes

Before defining the Stage 1 model formally, it is helpful to clarify why the VAR framework is useful in intensive longitudinal research. Like the random-intercept cross-lagged panel model (RI-CLPM; Hamaker et al., 2015), the present centered VAR separates stable person-specific levels from the dynamics of deviations around those levels. In the intensive longitudinal setting, this makes it possible to study how a person's current state depends on prior deviations in the same variable and in other variables over time. The model therefore captures both within-variable carryover—often called autoregressive persistence or inertia—and cross-variable lagged relations, in which one process predicts subsequent change in another. In

[2] We use the label MetaVAR because the Stage 1 model in this manuscript is a VAR fit separately to each individual. More generally, however, the broader framework implemented in the `metaDyn` R package requires only that each person contribute (a) a vector of estimated model features, $\hat{\boldsymbol{\theta}}_i$, and (b) an accompanying sampling covariance matrix, $\mathbf{V}_i^2$, so that first-stage uncertainty can be propagated into the population synthesis. In that sense, MetaVAR is the VAR-based instantiation of the broader `metaDyn` framework for dynamical systems.

that sense, the lagged structure also bears a loose resemblance to actor–partner logic: autoregressive effects capture how a variable predicts itself over time, whereas cross-lagged effects capture how one process predicts subsequent change in another (Gistelinck & Loeys, 2019, 2020). As such, the person-specific VAR provides a natural framework for studying short-term regulation, coupling, and spillover in multivariate psychological processes (Epskamp et al., 2018; Li et al., 2022).

More generally, let $\boldsymbol{\theta}_i \in \mathbb{R}^p$ denote the person-specific dynamic parameter vector for individual i, let $\hat{\boldsymbol{\theta}}_i$ denote its Stage 1 estimate, and let $\mathbf{V}_i^2 \in \mathbb{R}^{p \times p}$ denote the corresponding Stage 1 sampling covariance matrix. The contents of $\boldsymbol{\theta}_i$ are flexible: depending on the Stage 1 model, $\boldsymbol{\theta}_i$ may include autoregressive and cross-lagged coefficients, innovation variances and covariances, set point parameters, measurement-error variances, or other person-specific features of the dynamic system. The central requirement is that Stage 1 yield $\hat{\boldsymbol{\theta}}_i$ together with $\mathbf{V}_i^2$, or a good approximation to it, so that Stage 2 can propagate first-stage uncertainty rather than treat person-level estimates as error-free.

The diagonal elements of $\mathbf{V}_i^2$ quantify uncertainty about individual estimates, and its off-diagonal elements describe how their estimation errors vary together. This within-person estimation dependence differs from between-person covariation in the underlying dynamic parameters. An alternative Stage 1 estimator can be used if it supplies compatible parameter definitions and an aligned joint sampling covariance, rather than only a set of standard errors. We used the R package `fitVARMxID` because it provides these joint outputs for the centered VAR within the same likelihood-based workflow, including missing-data handling. It fits each person's model as a state-space model in `OpenMx` and evaluates the likelihood with the Kalman filter (Hunter, 2018), so unobserved occasions are handled within the likelihood rather than by deleting rows. This matters for lagged models: under listwise handling, one missing occasion removes both the row in which it is the outcome and the row in which it is a predictor. Equation-by-equation alternatives, such as ordinary least squares (e.g., the `vars` package, which does not accept missing values) or model-implied instrumental variable two-stage least squares (Bollen, 2019; Bollen et al., 2018), are useful estimators, but with listwise handling of incomplete series they can discard much of the data in sparse designs. The evaluation concerns this implementation; it does not establish equivalence to other VAR estimators or imply that every software package supplies the required covariance.

In this manuscript, Stage 1 fits a person-specific first-order VAR model. Let $\mathbf{s}_{i,t} \in \mathbb{R}^k$ denote the observed multivariate time series for person i at occasion $t = 1, \ldots, T$. The dynamic model follows a centered VAR specification (Asparouhov & Muthén, 2019; Hamaker & Grasman, 2015),

$$\mathbf{s}_{i,t} = \boldsymbol{\mu}_i + \mathbf{A}_i \left(\mathbf{s}_{i,t-1} - \boldsymbol{\mu}_i \right) + \boldsymbol{\zeta}_{i,t}, \quad \boldsymbol{\zeta}_{i,t} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Xi}_i), \tag{1}$$

where $\boldsymbol{\mu}_i$ is the person-specific set point, $\mathbf{A}_i$ collects autoregressive and cross-lagged effects governing the evolution of deviations from that set point, and $\boldsymbol{\Xi}_i$ is the innovation covariance matrix. Under stability or stationarity (e.g., all eigenvalues of $\mathbf{A}_i$ lie inside the unit circle) and mean-zero innovations, the process fluctuates around $\boldsymbol{\mu}_i$ such that $\mathbb{E}(\mathbf{s}_{i,t}) = \boldsymbol{\mu}_i$. For a bivariate process, the transition matrix may be written as

$$\mathbf{A}_i = \begin{pmatrix} a_{11,i} & a_{12,i} \\ a_{21,i} & a_{22,i} \end{pmatrix},$$

where the diagonal elements are autoregressive effects and the off-diagonal elements are cross-lagged effects. In the substantive language often used in affect-dynamics research, the autoregressive effects describe temporal persistence—often called *inertia*—in that they indicate how strongly deviations in a variable carry over to the next occasion. The cross-lagged effects describe lagged *coupling* between variables, indicating whether deviations in one process predict subsequent change in the other. We use these latter terms as descriptive labels for substantive interpretation, while retaining the standard VAR terminology in the formal model.

For the two-stage framework, Stage 1 contributes two outputs for each person: an estimated vector of dynamic features and a corresponding sampling covariance matrix that quantifies their estimation uncertainty. Stage 2 then treats those person-specific outputs as the objects to be synthesized at the population level.

For the present model, a convenient choice of person-specific parameter vector is

$$\boldsymbol{\theta}_i = (\boldsymbol{\mu}_i', \operatorname{vec}(\mathbf{A}_i)', \operatorname{vech}(\boldsymbol{\Xi}_i)')', \tag{2}$$

where $(\cdot)'$ denotes matrix transpose, $\operatorname{vec}(\cdot)$ stacks a matrix by columns, and $\operatorname{vech}(\cdot)$ stacks the lower triangle of a symmetric matrix, including the diagonal. The associated Stage 1 sampling covariance matrix,

$$\mathbf{V}_i^2 = \widehat{\mathbb{V}}\left[\hat{\boldsymbol{\theta}}_i\right],$$

can be obtained from the observed information matrix, a sandwich estimator, or a bootstrap, and is then passed to Stage 2. If some components are fixed or omitted in a given application, they are simply excluded from $\boldsymbol{\theta}_i$, and the dimension p is adjusted accordingly.

Stage 2 as Multivariate Mixed-Effects Meta-Analysis in SEM Form

At Stage 2, the objects being modeled are no longer the raw repeated measures, but the person-specific parameter estimates obtained from Stage 1. The SEM formulation is therefore defined on

the person-specific dynamic parameter system itself. This allows the second stage to model average dynamic features across persons, between-person heterogeneity in those features, and structured relations among them—for example, whether stronger inertia tends to co-occur with stronger cross-lagged coupling, or whether person-level covariates predict particular elements of the dynamic signature.

A convenient way to express this idea is through a measurement equation for the estimated person-specific parameters together with a structural model for the underlying person-specific parameter vector. Specifically, let

$$\hat{\boldsymbol{\theta}}_i = \boldsymbol{\theta}_i + \boldsymbol{\varepsilon}_i, \quad \boldsymbol{\varepsilon}_i \sim \mathcal{N}\left(\mathbf{0}, \mathbf{V}_i^2\right), \tag{3}$$

so that the observed Stage 1 estimate is treated as a noisy measurement of the latent person-specific dynamic parameter vector, with estimated sampling covariance matrix $\mathbf{V}_i^2$ treated as fixed in the Stage 2 likelihood. A general mixed-effects meta-analytic SEM for the latent parameter vector can then be written as (Cheung, 2008, 2013, 2014; Cheung & Cheung, 2016)

$$\boldsymbol{\theta}_i = \boldsymbol{\alpha}_i + \boldsymbol{\beta}\boldsymbol{\theta}_i + \boldsymbol{\Gamma}\mathbf{x}_i, \tag{4}$$

with person-specific latent parameter vectors

$$\boldsymbol{\alpha}_i = \boldsymbol{\alpha} + \boldsymbol{\upsilon}_i, \quad \boldsymbol{\upsilon}_i \sim \mathcal{N}\left(\mathbf{0}, \boldsymbol{\tau}^2\right). \tag{5}$$

Here, $\boldsymbol{\alpha}$ is the grand-mean vector of the dynamic parameter system, $\boldsymbol{\tau}^2$ is the between-person heterogeneity covariance matrix, and $\mathbf{x}_i$ collects person-level covariates. The matrices $\boldsymbol{\beta}$ and $\boldsymbol{\Gamma}$ allow researchers to specify structured relations and meta-regressions defined *on the parameter system itself.* In the present manuscript, however, our primary focus is on the random-effects synthesis case, which sets $\boldsymbol{\beta} = \mathbf{0}$ and omits covariates.

Assuming that $\mathbf{x}_i$ is non-stochastic and that $\mathbf{I} - \boldsymbol{\beta}$ is nonsingular, Equation 4 can be written in reduced form (Bollen, 1989) as

$$\boldsymbol{\theta}_i = \left(\mathbf{I} - \boldsymbol{\beta}\right)^{-1}\left(\boldsymbol{\alpha}_i + \boldsymbol{\Gamma}\mathbf{x}_i\right). \tag{6}$$

Combining this with Equation 3 yields the model-implied mean vector and covariance matrix of the observed Stage 1 estimates.[3]

$$\begin{aligned}\mathbb{E}\left[\hat{\boldsymbol{\theta}}_i \mid \mathbf{x}_i\right] &= (\mathbf{I}-\boldsymbol{\beta})^{-1}(\boldsymbol{\alpha}+\boldsymbol{\Gamma}\mathbf{x}_i),\\ \mathbb{V}\left[\hat{\boldsymbol{\theta}}_i \mid \mathbf{x}_i\right] &= (\mathbf{I}-\boldsymbol{\beta})^{-1}\boldsymbol{\tau}^2\left((\mathbf{I}-\boldsymbol{\beta})^{-1}\right)' + \mathbf{V}_i^2.\end{aligned} \tag{7}$$

In the special case $\boldsymbol{\beta}=\mathbf{0}$, the model reduces to a multivariate mixed-effects meta-analysis, or to multivariate meta-regression when $\boldsymbol{\Gamma}\neq\mathbf{0}$, with

$$\mathbb{E}\left[\hat{\boldsymbol{\theta}}_i \mid \mathbf{x}_i\right] = \boldsymbol{\alpha}+\boldsymbol{\Gamma}\mathbf{x}_i, \qquad \text{and} \qquad \mathbb{V}\left[\hat{\boldsymbol{\theta}}_i \mid \mathbf{x}_i\right] = \boldsymbol{\tau}^2+\mathbf{V}_i^2.$$

When no covariates are included and $\boldsymbol{\beta}=\mathbf{0}$, the model further reduces to a multivariate random-effects meta-analysis with

$$\mathbb{E}\left[\hat{\boldsymbol{\theta}}_i\right] = \boldsymbol{\alpha}, \qquad \text{and} \qquad \mathbb{V}\left[\hat{\boldsymbol{\theta}}_i\right] = \boldsymbol{\tau}^2+\mathbf{V}_i^2.$$

In this way, observed dispersion in $\hat{\boldsymbol{\theta}}_i$ is decomposed into (a) within-person estimation uncertainty carried by $\mathbf{V}_i^2$ and (b) genuine between-person heterogeneity carried by $\boldsymbol{\tau}^2$ (Pastor & Lazowski, 2018).

A common-effect (fixed-effects) synthesis is obtained by setting $\boldsymbol{\tau}^2=\mathbf{0}$, so that $\mathbb{V}[\hat{\boldsymbol{\theta}}_i]=\mathbf{V}_i^2$. Stage 1 sampling uncertainty, including covariance among estimates, is still propagated even when the underlying parameters are assumed identical across people. This differs from the uncertainty-uncorrected two-stage benchmark (Uncorr), which omits $\mathbf{V}_i^2$. Homogeneity can also be imposed on selected components while allowing other features to vary.

This system-level formulation is the key departure from edge-by-edge pooling. MetaVAR treats the full parameter vector as the unit of synthesis, thereby retaining the within-person sampling covariance among dynamic parameters and estimating a coherent between-person covariance structure for the dynamic system.

[3] In SEM software that supports *definition variables*, elements of the estimated, person-specific sampling covariance matrix $\mathbf{V}_i^2$ can be supplied as case-level data so that the model-implied covariance is recomputed for each individual (Lai & Hsiao, 2022; Mehta & Neale, 2005). This capability is central in meta-analytic applications because $\mathbf{V}_i^2$ is held fixed as a plug-in input, whereas common mixed-effects implementations (e.g., the `lmer` function) do not directly support fixing a case-specific sampling covariance matrix in the likelihood (Declercq et al., 2022). In the R package `OpenMx`, this is implemented by evaluating each individual's multivariate normal log-likelihood using that individual's model-implied mean vector and covariance matrix, which may depend on definition variables (Arbuckle, 1996; Enders, 2010; Neale et al., 2016). The `metaSEM` R package leverages this capability via `OpenMx` (Cheung, 2015); the `metaDyn` implementation of MetaVAR uses the same strategy to incorporate $\mathbf{V}_i^2$ and propagate Stage 1 uncertainty through Stage 2.

To describe heterogeneity, we report the conventional I^2 (Higgins & Thompson, 2002) separately for each dynamic parameter. For parameter p,

$$I_p^2 = \frac{\tau_p^2}{\tau_p^2 + \tilde{v}_p}, \tag{8}$$

where τ_p^2 is the pth diagonal element of the between-person covariance matrix $\boldsymbol{\tau}^2$, and $\tilde{v}_p$ is the typical Stage 1 sampling variance for that parameter,

$$\tilde{v}_p = \frac{(N-1)\sum_{i=1}^{N} w_{i,p}}{\left(\sum_{i=1}^{N} w_{i,p}\right)^2 - \sum_{i=1}^{N} w_{i,p}^2}, \qquad \text{and} \qquad w_{i,p} = \frac{1}{v_{i,p}^2}, \tag{9}$$

with $v_{i,p}^2$ denoting the pth diagonal element of $\mathbf{V}_i^2$ for individual i. Thus, I_p^2 gives the proportion of total variation in the estimated person-specific value of parameter p that is attributable to true between-person heterogeneity rather than Stage 1 estimation error. Values near zero indicate little heterogeneity relative to typical sampling variance, whereas values near one indicate that estimated heterogeneity is large relative to that variance. A near-unit value does not establish large absolute heterogeneity. We use I^2 descriptively in the empirical illustration; the simulation evaluates the means and variance–covariance components themselves, not recovery of I^2.

Model-Level Hypotheses and Constraints

Because Stage 2 is an SEM defined on the full parameter vector, MetaVAR is naturally suited to *model-level* hypotheses, that is, hypotheses that jointly constrain or relate multiple dynamic parameters rather than treating each coefficient as an isolated edge. This matters because many substantive predictions about regulation are multivariate by construction. For example, in an affective system one might ask whether inertia is comparable across negative affect and positive affect, whether cross-lagged coupling is asymmetric, whether stronger inertia tends to co-occur with stronger cross-affect coupling across persons, or whether particular elements of the dynamic signature predict later outcomes such as symptom severity or treatment response.

In MetaVAR, such questions can be expressed through standard SEM machinery applied to $\boldsymbol{\alpha}$, $\boldsymbol{\tau}^2$, $\boldsymbol{\beta}$, and $\boldsymbol{\Gamma}$. Equality constraints can be used to test whether average inertia parameters are comparable across constructs; structured relations in $\boldsymbol{\beta}$ can be used to model interrelations among components of the dynamic signature itself; and covariates or distal outcomes can be linked to selected elements of the parameter system through $\boldsymbol{\Gamma}$. For instance, one could test whether individuals with stronger negative-affect inertia also tend to show more negative coupling from negative affect to positive affect, or whether a distal

outcome is predicted by a composite of set point, inertia, and innovation parameters rather than by any single coefficient alone.

This has three advantages. First, inference remains system-aware: constraints and regressions are evaluated while retaining both the within-person sampling covariance $\mathbf{V}_i^2$ and the between-person heterogeneity structure $\boldsymbol{\tau}^2$. Second, hypotheses may be formulated not only for individual coefficients but also for functions of the dynamic system, such as coupling contrasts, symmetry constraints, or derived summaries of stability. Third, competing theory-level models can be compared using familiar SEM tools, including likelihood-based comparisons for nested models and information criteria for non-nested models.

In the present manuscript, however, our focus is on the core two-stage synthesis: person-specific dynamic parameters are estimated idiographically and then synthesized using multivariate random-effects meta-analysis, cast as meta-analytic SEM, with full propagation of each individual's Stage 1 sampling covariance matrix. Extensions involving covariates, distal outcomes, or richer second-stage structures are possible within the same template, but we treat them as future directions rather than as central demonstrations here.

Evaluation and Empirical Illustration

The remainder of the paper evaluates the operating characteristics of MetaVAR and demonstrates its applied utility. We first conduct a Monte Carlo simulation study that varies sample size and per-person series length and focuses on recovery of population-average dynamics, between-person heterogeneity, and uncertainty quantification when Stage 1 sampling covariance matrices are propagated into Stage 2. We then illustrate the end-to-end workflow using intensive longitudinal affect data. We conclude with a general discussion of the method's practical implications, limitations, and extensions.

Monte Carlo Simulation Study

We conducted a Monte Carlo simulation study to evaluate the operating characteristics of MetaVAR under a controlled bivariate affect-dynamics data-generating process (DGP) and to compare it with two practically important alternatives representing the main inferential strategies discussed in the Introduction: a one-step hierarchical estimator, implemented here as a Bayesian multilevel VAR benchmark (DSEM), and an uncertainty-uncorrected two-stage fit-many-then-summarize benchmark (Uncorr). The simulation examined recovery of (a) population-average dynamic parameters, (b) between-person heterogeneity in those parameters, and (c) uncertainty quantification across combinations of sample size (N) and per-person time-series length (T).

Because MetaVAR is specifically designed to propagate Stage 1 sampling covariance into a multivariate second-stage synthesis, we expected its clearest advantages to emerge for recovery of

between-person heterogeneity and for interval calibration for those heterogeneity parameters. At the same time, because the DGP and the Stage 1 estimation model were matched, the simulation represents a favorable, correctly specified setting for all methods.

Data-Generating Process

Data were generated using the `SimSSMVARIVary` function from the `simStateSpace` R package (v1.2.16; Pesigan et al., 2025). The data-generating model was the same centered first-order VAR model used in Stage 1 (Equation 1) for person $i = 1, \ldots, N$ and occasion $t = 1, \ldots, T$.

The person-specific set points were generated as

$$\boldsymbol{\mu}_i \sim \mathcal{N}\left(\begin{pmatrix} 2.87 \\ 2.04 \end{pmatrix}, \begin{pmatrix} 1.20 & 0.46 \\ 0.46 & 1.10 \end{pmatrix}\right), \tag{10}$$

and the person-specific transition matrices were generated by drawing

$$\operatorname{vec}(\mathbf{A}_i) \sim \mathcal{N}\left(\begin{pmatrix} 0.280 \\ -0.035 \\ -0.048 \\ 0.260 \end{pmatrix}, \begin{pmatrix} 0.017 & 0.005 & 0.001 & 0.008 \\ 0.005 & 0.008 & 0.001 & 0.006 \\ 0.001 & 0.001 & 0.001 & 0.002 \\ 0.008 & 0.006 & 0.002 & 0.026 \end{pmatrix}\right). \tag{11}$$

Here, $\operatorname{vec}(\cdot)$ denotes the vectorization operator that stacks the columns of a matrix into a single column vector. For a 2×2 matrix $\mathbf{A}_i = \left(a_{jk,i}\right)$, this implies $\operatorname{vec}(\mathbf{A}_i) = (a_{11,i},\, a_{21,i},\, a_{12,i},\, a_{22,i})'$.

Because the simulation allowed between-person heterogeneity only in the set points and transition parameters, the Stage 2 fixed-effect target can be written as

$$\boldsymbol{\alpha} = \left(\mathbb{E}[\boldsymbol{\mu}_i]', \mathbb{E}[\operatorname{vec}(\mathbf{A}_i)]'\right)'. \tag{12}$$

Likewise, the Stage 2 random-effects target is the between-person covariance matrix of $(\boldsymbol{\mu}_i', \operatorname{vec}(\mathbf{A}_i)')'$. Under the present DGP, this matrix is block diagonal because the set points and transition parameters were generated independently:

$$\boldsymbol{\tau}^2 = \begin{pmatrix} \boldsymbol{\Sigma}_\mu & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\Sigma}_A \end{pmatrix}. \tag{13}$$

This notation makes explicit that the fixed-effect targets correspond to $\boldsymbol{\alpha}$ and the heterogeneity targets correspond to $\boldsymbol{\tau}^2$.

The displayed transition covariance specifies the reference condition. We also used multipliers of zero and two on this covariance, leaving set-point heterogeneity unchanged. Thus, the zero condition makes all four transition coefficients identical across people and their between-person covariances zero; it does not make the entire person-specific system homogeneous. This condition examines whether estimation noise in person-specific models can make homogeneous parameters appear heterogeneous (Hoekstra et al., 2023). The doubled condition increases nominal transition variances and covariances, rather than doubling their standard deviations. Because unstable transition matrices were rejected, recovery is evaluated against the calibrated moments of the accepted population rather than automatically against the nominal normal-distribution moments. Exact-zero transition targets are analytic.

To ensure a stationary VAR process for each individual, generated transition matrices were retained only if all eigenvalues of $\mathbf{A}_i$ had moduli less than 1 (i.e., spectral radius < 1). In practice, this was implemented via rejection sampling using `simStateSpace`'s `SimBetaN` helper: candidate draws of $\text{vec}(\mathbf{A}_i)$ from the target multivariate normal distribution were repeatedly proposed until the corresponding $\mathbf{A}_i$ satisfied the stability constraint. Thus, the simulation targets a setting in which person-specific dynamics are heterogeneous but dynamically stable.

The innovation covariance matrix was held constant across persons and occasions,

$$\boldsymbol{\Xi} = \begin{pmatrix} 1.30 & 0.57 \\ 0.57 & 1.56 \end{pmatrix}, \tag{14}$$

so that the simulation isolates method performance for recovering heterogeneity in the set points and lagged dynamics. Fixing $\boldsymbol{\Xi}$ also simplifies interpretation because any between-person variation in estimated innovation parameters then reflects finite-sample estimation error rather than true heterogeneity in the DGP.

The parameter values were informed by prior empirical work on affective dynamics (Bringmann et al., 2013), but were not copied exactly. In particular, we parameterized the person-specific mean (set point) $\boldsymbol{\mu}_i$ rather than the intercept vector. This is substantively convenient because $\boldsymbol{\mu}_i$ directly represents each person's equilibrium level in the centered VAR parameterization, whereas the intercept in the uncentered form is a transformed quantity. The average transition matrix implies moderate positive autoregressive persistence for both variables and small negative cross-lagged effects, representing a stable bivariate regulatory system with weak reciprocal inhibitory coupling on average.

Design Factors

We primarily used a 3×3 factorial design crossing three sample sizes and three numbers of measurement occasions per individual:

$$N \in \{50, 100, 200\}, \qquad T \in \{50, 100, 200\}. \tag{15}$$

This yielded nine balanced sample-size/series-length patterns (Cases 1–9), where all individuals in a given replication shared the same series length T.

To induce heterogeneity in first-stage precision—and thus variability in the person-specific sampling covariance matrices used in Stage 2—we also included three unbalanced-T conditions (Cases 10–12). In these conditions, the sample size was fixed at $N \in \{50, 100, 200\}$ for Cases 10, 11, and 12, respectively, but the number of measurement occasions varied across individuals within the same replication. Specifically, for each individual i, the realized series length was $T_i \in \{50, 100, 200\}$, so that some individuals contributed shorter series and others longer series, producing heterogeneous $\mathbf{V}_i^2$ across persons. Analyses used each individual's realized T_i.

The unequal lengths were allocated cyclically in the order 50, 100, and 200, giving mean lengths of 115, 116, and 116.25 at $N = 50$, 100, and 200, respectively. Comparisons with balanced $T = 100$ therefore combine unequal precision with somewhat greater total information; they do not isolate an effect of unequal lengths at a fixed information budget.

Crossing the 12 sample-size/series-length patterns with the three transition-heterogeneity levels yielded 36 design cells. For each cell, $R = 1{,}000$ Monte Carlo replications were requested. In each replication, person-specific time series were generated under the DGP above, analyzed by each competing method, and summarized using the performance metrics described below.

Methods Compared

We compared three approaches representing the main inferential strategies discussed in the Introduction: (a) MetaVAR, a two-stage method with explicit propagation of first-stage uncertainty; (b) a one-step hierarchical benchmark, implemented here as Bayesian multilevel VAR (DSEM); and (c) an uncertainty-uncorrected two-stage fit-many-then-summarize benchmark (Uncorr), which summarizes person-specific estimates at Stage 2 without propagating their sampling covariance matrices. Across all methods, the inferential targets were the same: population-average dynamic parameters (fixed effects) and between-person heterogeneity in those parameters (random effects), as defined by the simulation DGP.

To facilitate comparability, all methods were fit to the same simulated datasets under each (N, T) condition, and summaries were mapped to a common parameterization whenever possible (i.e., set points,

VAR transition parameters, and between-person variability in those parameters). The comparison is therefore intended as a practical comparison of estimation strategies under a common target, not as a referendum on Bayesian versus frequentist inference.

MetaVAR

MetaVAR was implemented as a two-stage procedure using `fitVARMxID` (v1.0.6) for Stage 1 and `metaDyn` (v1.0.4) for Stage 2; both packages relied on `OpenMx` (v2.22.11; Hunter, 2018; Neale et al., 2016). In Stage 1, a person-specific VAR(1) model was fit separately for each individual to obtain (a) a vector of estimated dynamic parameters and (b) the corresponding estimated sampling covariance matrix. These outputs were then passed to Stage 2, where the person-level parameter vectors were synthesized using a multivariate random-effects meta-analytic model implemented in SEM form. Stage 2 uses these estimated sampling covariance matrices as fixed plug-in inputs, thereby propagating first-stage estimation uncertainty into the population-level likelihood without separately modeling uncertainty in the covariance estimates themselves.

A key inferential feature of MetaVAR is that it operates on the full parameter vector for each person rather than pooling parameters one at a time. Consequently, MetaVAR carries forward the within-person sampling covariance among estimated dynamic parameters into Stage 2 and estimates a coherent between-person random-effects covariance structure across the parameter system. In the simulation, we evaluated MetaVAR using model-based normal-theory confidence intervals at Stage 2.

The simulation synthesized six features (two set points and four transition coefficients), using their full sampling covariance, with separate two-feature and four-feature heterogeneity blocks. This estimates six means and 13 heterogeneity components, rather than an unstructured covariance for all nine Stage 1 features. The supplied sensitivity summaries also include robust Stage 2 intervals; these do not remove the nonstandard inference problem at a zero variance.

DSEM

The Bayesian multilevel VAR benchmark was implemented using `Mplus` 9.1 DSEM (Muthén & Muthén, 2017), with default and alternative-prior branches, two chains, and 40,000 iterations as specified in the simulation configuration. The default branch leaves priors to `Mplus`. For continuous-variable models, the documented defaults are diffuse normal priors for location parameters and improper inverse-Wishart priors $\text{IW}(\mathbf{0}, -d-1)$ for covariance blocks of dimension d (Asparouhov & Muthén, 2023). These are relatively uninformative on the specified parameter scales, but need not be innocuous for small heterogeneity components. The alternative branch uses $N(0, 25)$ for set-point means and $N(0, 0.25)$ for transition means, with the second argument denoting variance. Covariance priors are $\text{IW}(\mathbf{I}_2, 3)$ for the

set-point and innovation blocks and IW($0.004\mathbf{I}_4, 5$) for transition heterogeneity. The proper covariance priors and smaller transition scale provide a sensitivity analysis motivated by the influence of inverse-Wishart priors near small random-effect variances and the importance of prior sensitivity in SEM (Schuurman et al., 2016; van Erp et al., 2018). The numerical scales are study-specific choices rather than universal recommendations from those sources. Neither branch centers the priors on the generating values. Such priors would give DSEM population information that is unavailable in practice and that MetaVAR does not use, so the comparison would partly measure prior information rather than estimator performance. Data-informed priors are instead examined in the empirical illustration. The supplement details the prior specifications and their interpretation, and the simulation vignette provides example `Mplus` inputs and output.

Posterior quality was assessed using the saved simulation diagnostics: the rank-normalized potential scale reduction diagnostic $\widehat{R}$, bulk and tail effective sample sizes, and Monte Carlo standard errors relative to posterior standard deviations. A replication passed only when all required diagnostics were finite, $\widehat{R} \leq 1.01$, both effective sample sizes were at least 400, and relative Monte Carlo error was at most 0.05. We report passing counts for every design cell and prior branch, alongside requested and usable counts, and retain all usable runs in the primary performance summaries. Because these thresholds are demanding for multilevel models with many random-effect parameters, we also applied a moderate criterion as a sensitivity analysis: $\widehat{R} \leq 1.05$, both effective sample sizes at least 100, and relative Monte Carlo error at most 0.10.

Because posterior credible intervals and frequentist confidence intervals are not identical in interpretation, coverage comparisons should be understood as operating-characteristic summaries under this particular prior and Markov chain Monte Carlo (MCMC) configuration.

Uncertainty-Uncorrected Two-Stage Benchmark

As a baseline two-stage approach, the same saved Stage 1 fits used by MetaVAR supplied the same six person-specific point estimates to the uncertainty-uncorrected benchmark, Uncorr. It used the same set-point/transition block restrictions but set aside their sampling covariance matrices. Specifically, the second-stage analysis treated the Stage 1 point estimates as error-free and fit only the mean vector and covariance structure of the person-specific estimates using `OpenMx` (v2.22.11; Neale et al., 2016).

This benchmark reflects a common practical workflow in idiographic time-series research: fit models person-by-person, extract parameters of substantive interest, and then summarize those parameters across individuals. This general workflow is broadly consistent with common practice in person-specific network and multilevel VAR research, where individual models are often estimated separately and then compared or summarized across persons (Epskamp et al., 2018; Liu, 2017). This benchmark is included as

a transparent and practically common baseline two-stage workflow, not as an exhaustive representation of all possible two-stage alternatives. Its advantages are modularity, transparency, and computational simplicity. However, because the second-stage summaries do not account for the within-person sampling covariance of the Stage 1 estimates, the estimated between-person covariance of the Stage 1 estimates reflects both true heterogeneity and first-stage sampling variability.

Comparative expectations. Relative to the uncertainty-uncorrected two-stage benchmark, MetaVAR was expected to improve population-level inference by carrying forward person-specific sampling uncertainty into the second stage. Relative to DSEM, MetaVAR trades one-step joint estimation for a modular two-stage workflow that preserves idiographic estimation while supporting multivariate population synthesis. The simulation therefore evaluates not only point-estimate recovery, but also the inferential consequences of these distinct strategies.

Performance Metrics

We evaluated performance separately for the six population means in $\boldsymbol{\alpha}$ and the 13 freely estimated variances and covariances in $\boldsymbol{\tau}^2$. These heterogeneity parameters describe the population distribution of individual deviations, rather than the deviations themselves or I^2. Bias and root mean square error (RMSE) evaluate point recovery; coverage evaluates interval uncertainty; and detection evaluates sensitivity to nonzero targets. These metrics test the comparative expectations rather than presupposing an advantage from uncertainty propagation. All averages use the replications contributing a valid value to the relevant metric, which need not equal the 1,000 requested replications.

Point-Estimate Performance

For each target parameter θ, we computed Monte Carlo relative bias and RMSE across replications:

$$\begin{aligned} \text{Relative Bias}(\hat{\theta}) &= \frac{\bar{\hat{\theta}} - \theta}{\theta}, \\ \text{RMSE}(\hat{\theta}) &= \sqrt{R^{-1} \sum_{r=1}^{R} \left(\hat{\theta}_r - \theta\right)^2}, \\ \bar{\hat{\theta}} &= R^{-1} \sum_{r=1}^{R} \hat{\theta}_r. \end{aligned} \tag{16}$$

When summarizing the magnitude of relative bias without regard to direction, we used the absolute value of the relative bias: $\left|\text{Relative Bias}(\hat{\theta})\right|$.

For an exact-zero target, relative bias is undefined; we use bias $\bar{\hat{\theta}} - \theta$ and RMSE in the original units instead. In the formulas, R denotes the contributing count for that parameter and metric. For a reported proportion $\hat{p}$ based on R usable replications, its Monte Carlo standard error is $\sqrt{\hat{p}(1-\hat{p})/R}$.

Recovery of covariance components is interpreted on their own scales rather than pooled with population means into an overall error ranking.

Interval Performance (Empirical Coverage)

We evaluated uncertainty quantification using empirical coverage of nominal 95% interval estimates. For a given target parameter θ, empirical coverage was defined as

$$\widehat{\text{Coverage}}(\theta) = R^{-1} \sum_{r=1}^{R} \mathbb{I}(\theta \in \mathcal{I}_{r,.95}) \,, \tag{17}$$

where $\mathcal{I}_{r,.95}$ denotes the nominal 95% interval estimate obtained in replication r.

For MetaVAR and the uncertainty-uncorrected two-stage benchmark, $\mathcal{I}_{r,.95}$ was the nominal 95% confidence interval. For DSEM, it was the nominal 95% posterior credible interval. Although these interval types are not identical in interpretation, the empirical inclusion rate of nominal 95% intervals provides a practically useful basis for comparing uncertainty quantification across methods.

Coverage values near .95 indicate well-calibrated interval estimation. As a secondary benchmark, Bradley's liberal robustness criterion implies that, for nominal 95% intervals, empirical coverage between approximately .92 and .97 may be regarded as acceptable (Bradley, 1978). We therefore refer to this metric generically as empirical coverage of nominal 95% intervals.

Hypothesis-Testing Performance (Empirical Power)

For nonzero targets, we evaluated inferential sensitivity using empirical detection (power). For exact-zero transition-heterogeneity targets, we separately report interval exclusion of zero. For variances, boundary behavior and the support of the interval must be considered before interpreting exclusion as a calibrated test. For a given target parameter θ and nominal level α,

$$\widehat{\text{Power}}(\theta) = R^{-1} \sum_{r=1}^{R} \mathbb{I}(\text{reject } H_0 : \theta = 0 \text{ in replication } r) \,. \tag{18}$$

For MetaVAR and the uncertainty-uncorrected two-stage benchmark, rejection was based on whether the nominal 95% confidence interval excluded zero. For DSEM, we report the empirical detection rate based on whether the nominal 95% posterior credible interval excluded zero. Because several true effects in the DGP were small in magnitude—especially the cross-lagged effects and some heterogeneity components—empirical power is interpreted jointly with bias, RMSE, and coverage.

Monte Carlo Simulation Results

MetaVAR's principal advantage was the recovery of between-person transition heterogeneity across the range of sample sizes and series lengths. Relative to Uncorr, it reduced transition-variance RMSE in

nearly every target-by-condition comparison and transition-covariance RMSE in every comparison. At reference heterogeneity, its coverage averaged across the four transition variances was closer to nominal than both DSEM branches in every sample-size/series-length combination, including unequal lengths. The advantage narrowed as true heterogeneity increased and DSEM variance coverage improved. These findings establish the contribution across the design rather than at one favorable series length. Figures 1 and 2 show the overall and parameter-specific patterns; the supplement provides the complete condition-level evidence.

MetaVAR yielded usable results in 99.67% of the 36,000 requested replications; condition-specific counts and diagnostic details are reported in the supplement. All requested replications yielded usable summaries for Uncorr and both DSEM branches, but only 10,879 default-prior and 3,262 alternative-prior DSEM runs passed every posterior-quality criterion. Primary results retain all usable runs; the supplement reports the corresponding diagnostic counts. The failures were concentrated in the random-effect parameters. Averaged over conditions, the proportion of replications in which at least one random-effect parameter failed the strict criterion was .697 under default priors and .909 under alternative priors, against .034 and .242 for the fixed effects. Under the moderate criterion, the random-effect rates fell to .013 and .235, and the alternative-prior failures were concentrated in the 50-occasion conditions (see the supplement). Most strict-criterion failures therefore had diagnostics between the two sets of thresholds rather than severe nonconvergence; the exception was alternative-prior DSEM with short series, which also failed the moderate criterion in most replications (.714 on average).

Population-Average Parameters

Set-point means were recovered similarly by MetaVAR and Uncorr across the three heterogeneity levels, indicating that uncertainty correction made little practical difference for these well-recovered features. For transition means, MetaVAR generally improved RMSE and coverage over Uncorr, with the largest coverage gains for short series. As series length increased, both two-stage methods improved and their coverage differences narrowed. Increasing the number of participants did not substitute for longer individual series: residual idiographic bias became more consequential as population intervals narrowed, particularly with short series and little transition heterogeneity. DSEM generally provided better transition-mean recovery across the design. Thus, MetaVAR's uncertainty propagation improves two-stage synthesis but does not itself remove bias in the underlying person-specific point estimates.

Heterogeneity Parameters

The clearest advantage of MetaVAR emerged for transition heterogeneity. Across all three heterogeneity levels, it had lower RMSE than Uncorr for all transition-covariance targets and nearly all transition-variance targets. Under nonzero heterogeneity, longer series reduced Uncorr's upward distortion

of transition variation, but increasing the number of participants generally worsened its coverage: more precise synthesis did not remove the contamination of between-person variation by Stage 1 sampling error. MetaVAR's transition-variance coverage was substantially more stable across sample sizes and balanced or unequal lengths, averaging .928 at both reference and doubled heterogeneity, compared with .213 and .424 for Uncorr. Its transition-covariance coverage was also close to nominal across the nonzero conditions. The correction therefore mattered for both the amount of individual variation and the relationships among dynamic features. Set-point covariance summaries were much more similar across the two-stage methods.

The comparison with DSEM depended on the magnitude of heterogeneity. At reference heterogeneity, MetaVAR's average transition-variance coverage was closer to nominal under every sample-size/series-length pattern, not only with long series (Figure 2). Increasing true transition heterogeneity improved both DSEM branches, especially with longer series or more participants, bringing their variance coverage closer to MetaVAR's; the alternative-prior branch could match or exceed it in the better-informed doubled-heterogeneity conditions. MetaVAR also had lower transition-variance RMSE than default-prior DSEM in most target-by-condition comparisons at every heterogeneity level. The alternative priors narrowed that advantage and often yielded lower transition-covariance RMSE than MetaVAR. Covariance coverage under both DSEM branches was generally comparable to MetaVAR at nonzero heterogeneity. The strongest comparative case for MetaVAR is therefore stable inference for transition variation, especially when true variation is small, rather than superiority for every covariance or prior choice. These comparisons evaluate the full treatment of Stage 1 uncertainty; they do not isolate the incremental contribution of off-diagonal sampling covariances.

At zero transition heterogeneity, Uncorr still estimated appreciable positive variation. MetaVAR brought the estimates closer to zero, but this condition exposed a separate inference problem. Across cells and transition variances, its normal-theory intervals excluded zero at rates from .004 to .398; robust intervals gave .018 to .415. Neither interval variant supplied uniformly calibrated boundary inference. Averaged over transition variances, normal-interval exclusion decreased with longer series but increased with more participants, consistent with finite-series error becoming consequential as population inference grew more precise. DSEM variance intervals always excluded zero because their support was positive, which is not a calibrated test of the zero-variance null. Near-boundary estimates were retained when otherwise usable: the proportion of MetaVAR fits with a smallest heterogeneity-covariance eigenvalue below 10^{-8} ranged from .014 to .998 across cells, while the proportion with any variance at or below 10^{-6} ranged from zero to .026. Near rank deficiency can therefore occur without a near-zero diagonal variance. The supplement reports these rates by condition, separately from inadmissibility.

Testing an exact zero asks more than most applications need, because a variance can be negligible without being zero. We therefore also evaluated practical homogeneity of the four transition variances against a tolerance δ, following the logic of equivalence testing (Lakens et al., 2018): an upper 95% limit at or below δ supports practical homogeneity, whereas a lower limit above δ indicates heterogeneity beyond the tolerance. The primary tolerance was $\delta = .0009$, a between-person standard deviation of .03, the benchmark for a small cross-lagged effect (Orth et al., 2024); $\delta = .0004$ and $\delta = .0025$ (standard deviations of .02 and .05) served as sensitivity values. Across the 12 zero-heterogeneity cells, MetaVAR's normal-theory upper limits fell at or below δ for .093 of the transition-variance estimates on average (robust intervals, .118), rising with series length to .298 when $N = 200$ and $T = 200$. Its lower limits exceeded δ for .023 (robust, .029), although this rate, averaged over the four transition variances, reached .165 with 200 participants and 50 occasions. Uncorr never supported practical homogeneity, and its lower limits always exceeded δ. Neither DSEM branch supported practical homogeneity at this tolerance; their lower limits exceeded δ for .761 of the estimates under default priors and .025 under alternative priors. At the larger tolerance, alternative-prior DSEM supported practical homogeneity for .262 of the estimates, compared with .364 for MetaVAR (see the supplement). With a tolerance, MetaVAR seldom mistook homogeneous transitions for heterogeneous ones except with many participants and short series, but its intervals were usually too wide to establish homogeneity unless individual series were long.

For small nonzero targets, detection depended on both available information and the magnitude of heterogeneity. Averaged over transition covariances and participant sample sizes, MetaVAR's detection increased with series length at both nonzero heterogeneity levels and was higher under doubled heterogeneity, while covariance coverage remained close to nominal. Default-prior DSEM showed a similar detection pattern; alternative priors generally increased detection, with accompanying target-specific differences in RMSE and coverage. Uncorr could have higher detection for short series despite poor coverage, and that apparent advantage diminished or reversed as series length and true heterogeneity increased. MetaVAR thus offered increasing sensitivity with more informative individual series without relying on the spurious heterogeneity that inflated uncorrected detection. The supplement reports nonzero detection separately from interval exclusion at exact-zero targets.

Larger participant samples and longer individual series played different roles. More participants sharpened inference about the population distribution, whereas longer series improved the person-specific estimates entering that inference. MetaVAR's weakest nonzero transition-variance coverage occurred when short series were combined with the largest participant sample, illustrating why a large number of people cannot compensate fully for limited within-person information. Across the remaining patterns, its heterogeneity coverage was comparatively stable. Unequal lengths extended this evaluation to varying

person-specific precision: at matched participant size and heterogeneity, MetaVAR and both DSEM branches maintained average coverage close to their balanced-length counterparts, whereas Uncorr generally deteriorated. MetaVAR's heterogeneity RMSE improved in 80 of 117 comparisons with balanced $T = 100$, versus 10 of 117 for Uncorr. This supports the practical value of carrying person-specific precision into synthesis. The unequal condition also contained more observations on average, so it does not isolate unequal precision at a fixed information budget.

A dedicated benchmark on the same simulated dataset ($N = 200$, $T = 200$, reference heterogeneity; 100 repetitions per method) demonstrated a substantial computational advantage over joint estimation. Mean elapsed times were 80.33 seconds for MetaVAR, 9.85 seconds for Uncorr, and 745.97 seconds for DSEM under default priors. Both two-stage timings included Stage 1 fitting and Stage 2 synthesis. DSEM therefore took approximately 9.3 times as long as MetaVAR in this configuration. Uncorr was fastest, but its lower cost came with the heterogeneity-recovery problems described above. The benchmark used `Mplus` 9.1 on Linux and parallel person-specific estimation; the benchmark vignette documents hardware, software, and allocation settings. These timings measure execution under the specified settings, not time to equally converged solutions or a universal speed ratio. Independent Stage 1 fits can also be reused across different Stage 2 models, extending the practical value of the two-stage workflow.

Taken together, these results support MetaVAR as an efficient approach to inference on between-person heterogeneity in dynamic systems. Propagating Stage 1 uncertainty markedly improved transition-heterogeneity recovery over Uncorr, and transition-variance coverage compared favorably with both DSEM prior branches. DSEM's stronger transition-mean recovery remains valuable when average dynamics are the primary target. Short-series bias and variance-boundary inference define important limits to MetaVAR's use.

Monte Carlo Simulation Discussion

The simulation supports MetaVAR as an efficient approach to learning how dynamic systems differ across people. Its benefit over an uncorrected workflow was evident across the evaluated sample sizes, balanced and unequal series lengths, and heterogeneity levels: transition-covariance RMSE was consistently lower, and transition-variance RMSE was lower in nearly every comparison. With nonzero heterogeneity, its coverage remained substantially more stable than Uncorr's as participant sample size changed. The advantage over DSEM was clearest for transition-variance coverage at reference heterogeneity and persisted across every sample-size/series-length pattern. Larger heterogeneity and greater information made the DSEM alternatives more competitive, while DSEM remained stronger for population-average transitions. Set-point summaries were often similar across the two-stage methods. These patterns identify MetaVAR's practical role: preserving idiographic estimation while separating uncertainty in those estimates from

between-person variation, with a substantial computational saving relative to the evaluated joint implementation.

The zero-heterogeneity condition also clarifies the distinction between specifying a common effect and estimating whether effects vary. When homogeneity is substantively justified, MetaVAR can fix the relevant heterogeneity components to zero while retaining Stage 1 uncertainty. The simulation instead estimated those components, exposing the limitations of normal and robust Wald intervals at the boundary. Performance of a prespecified common-effect model was not evaluated here.

Empirical Illustration: Meta-Analysis of Affect Dynamics

We illustrate the proposed MetaVAR workflow using intensive longitudinal data from the Affective Dynamics and Individual Differences study (ADID; Emotions and Dynamic Systems Laboratory, 2010), which was designed to examine how emotion dynamics unfold in everyday life and how these dynamics differ across individuals. The ADID EMA protocol sampled participants five times per day for approximately 30 days (four daytime prompts plus an evening report), yielding up to 150 measurement occasions per participant in the raw data. ADID has contributed to multiple substantive and methodological studies, including work on regime-switching dynamics, measurement/trait–state structure, network and VAR-based modeling, outlier-robust state-space modeling, and missing-data methods (Chow & Zhang, 2013; Gates et al., 2023; Guo et al., 2012; Hutton & Chow, 2014; Li et al., 2022, 2025; Oh et al., 2025; Ou et al., 2023; Park et al., 2020; You et al., 2020).

This setting closely matches the goal of MetaVAR: estimate person-specific dynamic signatures in a first stage and then synthesize them at the population level in a second stage while propagating each individual's Stage 1 sampling uncertainty. In the present illustration, we focus on within-person dynamics between negative affect (NA) and positive affect (PA). Specifically, we meta-analyze the full person-specific parameter system consisting of (a) affective set points, (b) temporal dynamics encoded by the VAR(1) transition matrix—autoregressive (AR) inertia and cross-lagged coupling—and (c) the innovation covariance matrix, which captures within-occasion co-fluctuation of unexpected changes after accounting for lagged dynamics.

Substantively, NA and PA are useful targets because emotion-dynamics theories treat affect as a self-regulating system characterized by a typical level (often described as a set point), temporal persistence from one occasion to the next (often described as inertia), and possible cross-lagged coupling between affect dimensions that may reflect carryover or regulatory spillover (Houben et al., 2015; Kuppens, Allen, & Sheeber, 2010; Kuppens, Oravecz, & Tuerlinckx, 2010). MetaVAR treats heterogeneity in these features—including heterogeneity in the innovation covariance structure—as a scientific target, providing

population-average inference while quantifying between-person variation and co-variation across the dynamic system.

Measures

Momentary affect in ADID was assessed using adjective ratings drawn from established affect instruments, including items from the Positive Affect and Negative Affect Schedule (PANAS; Watson et al., 1988) and circumplex-content items used to broaden coverage of activation levels (Russell, 1980). Participants rated how often they had experienced each affect descriptor since the previous assessment on an ordinal Likert-type scale, as implemented in the ADID daily diary surveys.

Following prior ADID applications that regularized the irregular EMA observations to equally spaced intervals for discrete-time modeling and applied within-person linear detrending, we constructed per-occasion NA and PA composites by averaging the available NA and PA items (Park et al., 2020; You et al., 2020). We then rounded the recorded date–time stamps to the nearest hour and inserted missing values for hours with no report to obtain an approximately hourly series for the present VAR analysis. Linear detrending preserved each person's observed mean. The retained sample contributed 146,677 regularized grid rows, with 24,677 observed NA values and 24,674 observed PA values; the grid-row count is not a count of observed affect reports. These steps align the data with the discrete-time VAR framework used here, so the lagged coefficients are interpreted as approximately hourly dependencies in detrended affect rather than as features of the full irregularly spaced raw month-long trajectory.

Results

Stage 1 estimated person-specific NA–PA dynamics for each participant, and Stage 2 synthesized these individual dynamic signatures to obtain (a) population-average dynamics, (b) between-person heterogeneity in those dynamics, and (c) systematic co-variation among elements of the parameter system.

Stage 1: Person-Specific VAR

We fit the person-specific VAR(1) model in Equation 1 separately for each participant using the `fitVARMxID` package (v1.0.6), estimating all dynamic parameters as person-specific: set points, autoregressive coefficients, cross-lagged coefficients, and innovation covariance parameters. The analysis included 212 participants.

Stage 2: Multivariate Random-Effects Meta-Analysis of the Dynamic Parameter System

Stage 2 synthesized the person-specific VAR(1) parameter systems using a multivariate random-effects meta-analysis with the `metaDyn` package (v1.0.4). The meta-analytic mean vector $\boldsymbol{\alpha}$ contained nine elements, ordered as the two set points ($\mu_{\mathrm{NA}}, \mu_{\mathrm{PA}}$), the four VAR transition coefficients

$(a_{11}, a_{21}, a_{12}, a_{22})$, and the three unique elements of the innovation covariance matrix $\boldsymbol{\Xi}$ for NA and PA $(\xi_{11}, \xi_{21}, \xi_{22})$.

The Stage 2 inputs included both the nine-element estimate vector and its aligned 9×9 sampling covariance matrix for each participant. Thus, the innovation covariance $\boldsymbol{\Xi}_i$, which describes within-occasion affective shocks, is distinct from the sampling covariance $\mathbf{V}_i^2$, which describes uncertainty about all nine estimated features. The empirical specification estimates nine population means and an unstructured heterogeneity matrix with 45 distinct variance–covariance elements. The saved Stage 2 vectors and sampling covariances were verified against the Stage 1 fits. Numerical conditioning of the sampling covariance matrices and the common-input comparison are documented in the supplement.

Population-average dynamics ($\boldsymbol{\alpha}$). The population-average set points indicated higher typical PA than NA (2.573 versus 1.442; Table 1). Both affect dimensions showed positive inertia: $a_{11} = 0.383$ for NA and $a_{22} = 0.329$ for PA. Both cross-lagged effects were negative on average, with $a_{21} = -0.095$, 95% CI $[-0.161, -0.029]$, and $a_{12} = -0.039$, 95% CI $[-0.075, -0.002]$. These estimates describe reciprocal negative hourly dependencies, with weaker evidence for the PA→NA effect. They do not establish mechanistic inhibition or a difference between the two effects.

The average innovation variance was larger for PA than for NA (0.0623 versus 0.0318), and the innovation covariance was negative (−0.0079). Thus, after accounting for lagged dynamics, unexpected within-occasion changes in NA and PA tended to move in opposite directions on average.

Between-person heterogeneity and co-variation ($\boldsymbol{\tau}^2$). The estimated heterogeneity matrix indicated variation across the parameter system. Between-person standard deviations were 0.366 and 0.467 for NA and PA set points, 0.436 and 0.435 for their inertia, and 0.418 and 0.241 for the NA→PA and PA→NA coefficients. Table 1 gives all nine standard deviations. A likelihood ratio test of whether the transition coefficients varied across people compared this model with one in which the four transition variances, their six covariances, and their 20 covariances with the other features were fixed at zero (30 constraints). Because the null hypothesis places variances on the boundary of the parameter space, the statistic does not follow the usual χ^2 distribution (Self & Liang, 1987; Stoel et al., 2006; Stram & Lee, 1994). We referred it to a chi-bar-square distribution, whose mixing weights were obtained by simulation from the Fisher information at the null estimates (Shapiro, 1988), and to the conservative $\chi^2(30)$ distribution. The statistic was 17,342, far above both critical values (36.55 and 43.77; $p < .001$), so the hypothesis of homogeneous transitions was rejected.

The covariance structure also revealed associations among dynamic features. The estimated covariance of NA inertia with NA→PA coupling was negative (−0.0304), but its 95% interval [−0.0611, 0.0002] narrowly included zero. The covariance of PA inertia with PA→NA coupling was −0.0293, 95% CI

[−0.0467, −0.0119]. In contrast, the estimated positive covariance between the NA set point and NA inertia was 0.0204, with a 95% interval [−0.0026, 0.0434] that included zero. These associations characterize between-person differences in dynamic signatures; they are not additional within-person temporal effects.

The uncorrected analysis used exactly the same Stage 1 point estimates. Set-point means were almost identical, but corrected inertia means were higher than uncorrected means (0.383 versus 0.343 for NA; 0.329 versus 0.286 for PA). The uncorrected between-person SD for NA→PA coupling was 0.567, compared with MetaVAR's 0.418. Innovation-feature dispersion was also larger in Uncorr. The adjustment was not a uniform reduction of every variance: PA-inertia SD was slightly higher in MetaVAR (0.435 versus 0.429). A joint likelihood with person-specific precision can change both location and covariance estimates. This comparison demonstrates the empirical consequences of uncertainty propagation, without establishing accuracy against unknown population values.

Heterogeneity index (I^2). The I^2 values ranged from .933 to approximately 1.000 (Table 1). These indices describe heterogeneity relative to typical Stage 1 sampling variance, rather than its absolute substantive magnitude. Greater Stage 1 precision can yield a large I^2 even for modest between-person variation; interpretation therefore requires the native-scale standard deviations alongside the index. Neither the index nor agreement between methods establishes estimator accuracy when the population parameters are unknown.

Empirical Example Discussion

This illustration shows how MetaVAR can be used to estimate idiographic affect-dynamics parameters from intensive longitudinal data and then synthesize those person-specific signatures into population-level inferences while quantifying heterogeneity. In ADID, the approach yielded interpretable population-average dynamics, strong evidence of between-person heterogeneity in both set points and lagged dependencies, and substantively meaningful co-variation among elements of the parameter system (Houben et al., 2015). In addition to the familiar temporal dependencies encoded in the VAR transition matrix, MetaVAR also supports population inference on the innovation structure, which captures within-occasion co-fluctuation in unexpected affective shifts after accounting for lagged dynamics.

Population-Average NA–PA Dynamics

At the population level, participants reported higher typical PA than NA, consistent with work showing that average affective tone in community samples tends to be more positive than negative (Houben et al., 2015; Watson et al., 1988). Both NA and PA showed positive inertia, with AR(NA) slightly larger than AR(PA), suggesting that momentary elevations in NA persisted somewhat more strongly than elevations in PA at the hourly scale used here (Houben et al., 2015; Kuppens, Oravecz, & Tuerlinckx, 2010).

Cross-lagged coupling was negative in both directions: higher NA predicted lower subsequent PA, and higher PA predicted lower subsequent NA. Interpreted as a descriptive bivariate regulatory system, these results are broadly consistent with reciprocal inhibition or spillover processes discussed in affect-dynamics theory (Kuppens, Allen, & Sheeber, 2010; Kuppens, Oravecz, & Tuerlinckx, 2010). At the same time, these coefficients should be interpreted as average short-term dependencies at a specific time scale (hourly, after detrending), not as mechanistic causal effects (Asparouhov & Muthén, 2019; Hamaker & Grasman, 2015).

These findings are partly consistent with earlier ADID analyses, which likewise supported lagged affective dependencies but suggested a more asymmetric cross-lagged pattern. In particular, You et al. (2020), using a coarser equally spaced aggregation of the irregular EMA observations together with within-person linear detrending, suggested stronger evidence for a PA→NA lagged effect than for the reverse NA→PA effect. The present MetaVAR analysis instead suggests negative average cross-lagged effects in both directions. One plausible reason is timescale: the present study regularized the series to an approximately hourly grid, whereas the earlier state-space analysis used a coarser temporal aggregation. Another difference is inferential target. Earlier ADID analyses were not designed to estimate a full random-effects covariance structure for the person-specific dynamic parameter system. MetaVAR therefore adds information those analyses could not directly target, including between-person heterogeneity in inertia and cross-lagged coupling as well as covariation among these dynamic features across individuals (Park et al., 2020; You et al., 2020).

The population-average innovation covariance matrix further indicated that, after accounting for lagged dynamics, within-occasion unexpected shifts in NA and PA tended to move in opposite directions on average. The larger innovation variance for PA than for NA suggests that short-term unexplained fluctuations were greater for PA in this dataset.

Heterogeneity as a Substantive Finding, Not a Nuisance

Between-person variation in set points, lagged dynamics, and innovation parameters is a substantive target of the synthesis. Its magnitude should be assessed through $\boldsymbol{\tau}^2$ on the scale of each feature, with I^2 providing complementary information about relative precision (Higgins & Thompson, 2002). More informative individual series reduce typical sampling variance and can increase I^2 without increasing absolute heterogeneity. Off-diagonal elements of $\boldsymbol{\tau}^2$ address a separate question: how differences in one dynamic feature co-occur with differences in another.

The covariance results illustrate this point. The intervals for NA set-point–inertia covariation and NA inertia–coupling covariation included zero. Stronger PA inertia was associated with more negative PA→NA coupling, indicating that temporal persistence and cross-affect coupling need not vary

independently across people. Co-variation among the innovation parameters further suggests meaningful between-person differences in the magnitude and structure of within-occasion affective shocks.

Practical and Methodological Considerations

The DSEM comparison also illustrates the practical computational demands of joint estimation. The default-prior fit required 36 hours, 2 minutes, and 54 seconds, and the alternative-prior fit, whose priors were scaled to the affect data rather than taken from the simulation (see the supplement), required 35 hours, 35 minutes, and 58 seconds in `Mplus` 9.1. Despite roughly a day and a half of computation per fit, neither achieved adequate posterior convergence. Both supplied 212-participant fits terminated normally, but their final reported potential scale reduction values were 1.451 under default priors and 1.366 under alternative priors, with the largest discrepancy associated with the between-person variance of PA inertia. The saved chains likewise showed substantial disagreement: rank-normalized $\widehat{R}$ for the PA-inertia variance was 1.442 and 1.462, with bulk effective sample sizes below eight in both branches. Inertia means also failed the posterior-quality screen. These fits do not support a settled empirical comparison of the full parameter system. Their diagnostic limitations are summarized in the supplement; normal termination is not treated as evidence of adequate convergence. Empirical differences between methods, even with satisfactory diagnostics, would remain differences between estimates rather than evidence of accuracy against known truth.

To give DSEM a further chance, we refitted it twice with data-informed priors centered on the MetaVAR estimates, following the recommendation to include a data-based prior in sensitivity analyses of inverse-Wishart specifications (Schuurman et al., 2016). Each mean had a normal prior with the MetaVAR estimate as its mean and the MetaVAR sampling variance as its variance. Each covariance block had an inverse-Wishart prior with $212 + d + 1$ degrees of freedom, whose mean was the MetaVAR between-person covariance matrix (full version) or its diagonal (diagonal version), so that the prior carried the weight of 212 persons. The full version converged (final potential scale reduction 1.023; rank-normalized $\widehat{R} \leq 1.03$ for all 28 parameters), and the diagonal version nearly did (1.050; $\widehat{R}$ above 1.05 only for the NA-inertia mean and variance), in 35 hours, 19 minutes and 35 hours, 28 minutes. Set-point means and variances were nearly identical across all four fits and MetaVAR. With weak priors, inertia means were higher (.467 and .452 for NA and PA, versus .383 and .329 for MetaVAR) and transition variances smaller (.015 and .003 for the two cross-lagged coefficients under default priors and .020 and .008 under alternative priors, versus .175 and .058). With data-informed priors, the means moved close to the MetaVAR values (.401 and .348), and the variances moved most of the way (.127 and .041). The covariances among transition coefficients followed the prior: they approached the MetaVAR values under the full version but stayed near zero under the diagonal version, as in the weak-prior fits.

These results suggest that the data alone only weakly identify between-person heterogeneity in the dynamics, especially its covariance structure, so DSEM estimates of these components depend on the prior. The data-informed fits are not independent confirmation of MetaVAR. Their priors reuse the same data at a weight equal to the sample, so their agreement with MetaVAR and their narrower intervals (posterior standard deviations of the set-point means fell by about 30%) arise partly by construction. The supplement reports all four fits.

The hourly grid offers one plausible explanation for these difficulties. Approximately 83.2% of grid entries were unobserved for each affect composite. Although the grid was constructed in R rather than with `Mplus`'s `TINTERVAL` command, it creates the same need to account for unobserved intermediate occasions. Asparouhov and Muthén (2024a) explain that fine discretization can enlarge the missing-data augmentation problem and slow MCMC convergence. Continuous-time modeling can avoid this grid expansion, but evaluating that alternative is beyond the present study. We did not vary the grid or refit the models, so its contribution to the observed mixing problems remains a plausible mechanism rather than an identified cause. MetaVAR nevertheless supplied a usable population synthesis in an application where joint models with weak priors remained inconclusive after prolonged computation and converged only with priors built from the same data.

Several features of the analysis highlight common trade-offs in applied dynamic modeling. First, we imposed regular spacing by rounding responses to the nearest hour and inserting missing values for unobserved hours, and we applied within-person linear detrending to better align the data with a stationary VAR(1) approximation. These choices improve feasibility and interpretability within the present discrete-time framework, but they also mean that the empirical target is a detrended, approximately hourly affective process rather than the full irregularly spaced raw EMA record, shifting interpretation toward shorter-term fluctuations rather than slower drift across the month-long protocol (Guo et al., 2012; Li et al., 2022). The results are also conditional on successful idiographic estimation, which may limit generalization to participants whose models could not be estimated reliably. Second, NA and PA composites were constructed from ordinal item ratings and treated as approximately continuous. Future work could examine whether latent measurement models or item-level analyses alter the estimated dynamic signature or its heterogeneity (Park et al., 2020; Watson et al., 1988).

Implications and Future Directions

The population means and heterogeneity covariances reported here summarize the distribution of person-specific parameters. They are not replacements for the unpooled Stage 1 estimates, and the illustration does not establish improved accuracy of person-level estimates through shrinkage. Evaluating such predictions would require a separate analysis.

Overall, the illustration supports the value of treating affect dynamics as person-specific signatures that can be meaningfully compared and synthesized (Fisher et al., 2017; Houben et al., 2015). Several extensions follow naturally from the same Stage 2 SEM template but are not pursued here, including covariate moderation of the parameter system and links between dynamic signatures and distal outcomes under full Stage 1 error propagation. Additional directions include allowing nonstationarity (e.g., regime switching or time-varying parameters; Chow & Zhang, 2013; Hutton & Chow, 2014), extending Stage 1 to continuous-time formulations to better handle irregular sampling without discretization (Park et al., 2025), and targeting multivariate constraints or derived functionals—such as stability summaries, impulse-response metrics, or innovation-based summaries—as higher-level quantities for synthesis. A natural next step is to place person-level emotion-regulation markers and related covariates on the Stage 2 parameter system to explain why individuals differ in set points, inertia, coupling, and innovation structure.

General Discussion and Future Directions

The central aim of this paper was to develop a principled way to move from person-specific dynamic models to population-level inference without collapsing the distinction between within-person and between-person structure. MetaVAR addresses this aim by combining idiographic estimation in Stage 1 with multivariate meta-analytic structural equation modeling in Stage 2. In doing so, it brings together two lines of work that are usually kept separate: recent two-stage idiographic dynamic-modeling approaches that treat heterogeneity as a target of inference rather than a nuisance (Lee & Gates, 2024; Zhang et al., 2025), and the MASEM literature, which provides machinery for modeling multivariate effect sizes, sampling covariances, heterogeneity, and structured parameter relations within a single SEM framework (Cheung, 2008, 2014; Cheung & Cheung, 2016; Jak & Cheung, 2020). In this sense, MetaVAR is not simply a new way to average person-specific coefficients. It is a system-level synthesis framework in which persons play the role of studies and each person's dynamic signature is treated as a multivariate effect-size vector. More broadly, the present contribution complements earlier comparisons of person-specific and multilevel autoregressive modeling (Liu, 2017) by showing that the relevant methodological question is not only whether within-person dynamics should be estimated idiographically or hierarchically, but also how idiographic estimates should be synthesized once obtained if the goal is valid population-level inference on heterogeneity and cross-parameter co-variation.

This positioning also clarifies where MetaVAR fits in the broader methodological landscape. MetaVAR is not intended as a universal replacement for one-step hierarchical modeling. When the main scientific target is the population-average dynamic structure, or when partial pooling is itself substantively desirable, one-step multilevel models remain highly attractive. The simulations make this distinction concrete: DSEM generally recovered transition means better, while MetaVAR provided more reliable

transition-variance coverage at reference heterogeneity under both DSEM prior choices. At the same time, that comparison should be interpreted as a practical benchmark based on a common `Mplus` implementation with default and explicitly specified alternative priors, not as a definitive evaluation of all Bayesian multilevel VAR specifications. In one-step Bayesian estimation, recovery of variance–covariance components can depend materially on prior specification for random-effects covariance structures, especially when those parameters are small or weakly identified (Asparouhov & Muthén, 2024b; Li et al., 2022; Schuurman et al., 2016; van Erp et al., 2018). The present results therefore speak most directly to the practical trade-offs among these particular implementations rather than to a general Bayesian–frequentist contrast.

MetaVAR's clearest contribution is to valid two-stage multivariate inference when the scientific target includes between-person heterogeneity and cross-parameter co-variation in the dynamic system. By carrying forward each person's Stage 1 sampling covariance matrix $\mathbf{V}_i^2$ rather than treating person-specific estimates as error-free, MetaVAR separates first-stage estimation noise from genuine between-person differences and retains dependence among estimated dynamic parameters. This matters because, in multivariate meta-analysis more generally, ignoring within-study covariance can distort estimation and inference, particularly when the goal is to synthesize multiple correlated outcomes jointly (Berkey et al., 1998; Jackson et al., 2011; Riley, 2009; van Houwelingen et al., 2002). The simulation results were consistent with that logic. MetaVAR substantially improved transition-heterogeneity recovery relative to the uncorrected workflow across sample sizes and series-length patterns, including unequal lengths, and maintained average transition-variance coverage of .928 at both nonzero heterogeneity levels. This was particularly advantageous when transition variation was small, where both DSEM prior branches undercovered. Set-point summaries were often similar, and DSEM could provide comparable covariance recovery, depending on the prior and target. The zero-heterogeneity results also show that uncertainty propagation does not make ordinary Wald intervals reliable at a variance boundary. That distinction is substantively important because many questions in intensive longitudinal research concern not only what dynamics look like on average, but also how and why people differ in those dynamics.

The empirical illustration reinforces this point. In the ADID data, MetaVAR estimated interpretable population-average affect dynamics while also revealing substantial heterogeneity in set points, autoregressive persistence, cross-lagged coupling, and innovation structure. More importantly, the off-diagonal elements of $\boldsymbol{\tau}^2$ showed that these features did not vary independently across people: dynamic characteristics such as inertia, coupling, and innovation structure co-varied in systematic ways. This extends earlier ADID analyses that were informative about average lagged dynamics and preprocessing choices, but were not designed to estimate a full random-effects covariance structure for the person-specific parameter system (Park et al., 2020; You et al., 2020). At the same time, the present empirical target

should be interpreted carefully. Because the series were regularized to approximately hourly bins and detrended before estimation, the resulting parameters describe a short-term, detrended, approximately hourly affective process rather than the full irregularly spaced raw month-long EMA trajectory.

Computational efficiency strengthens this inferential contribution. The dedicated benchmark included both stages and required about one ninth of the DSEM execution time under the evaluated configuration. Separating independent person-specific fits from population synthesis also allows researchers to reuse Stage 1 estimates when examining different Stage 2 hypotheses. The empirical illustration shows why this flexibility matters in practice: each of the four DSEM fits required about 35 to 36 hours, and only the fits with priors built from the same data converged, while MetaVAR yielded a usable synthesis of the nine-feature parameter system. These findings support MetaVAR as a practical alternative when joint estimation is expensive or difficult to fit. They establish a computational advantage for the evaluated workflow, while statistical performance in larger-dimensional systems remains an open question.

Several limitations qualify these conclusions. The simulation considered a bivariate data-generating process with one mean dynamic system, three levels of transition heterogeneity, fixed innovation covariance across persons, and correct model specification. Varying transition heterogeneity broadens this evaluation but does not establish performance for different mean systems, heterogeneous innovation processes, or larger numbers of variables. The empirical illustration relied on observed composites, discretization of irregular measurements, and a stationary VAR(1) approximation after detrending. The simulation also evaluated DSEM under only two prior specifications. Data-based inverse-Wishart priors, which performed best in a comparison of inverse-Wishart specifications for multilevel autoregressive models (Schuurman et al., 2016), and half-t priors on the random-effect standard deviations (Gelman, 2006) were not evaluated. Because DSEM's recovery of small transition variances depended on the prior, other specifications could narrow or widen the differences reported here; a simulation that includes them would address this question. Accordingly, the present results speak to performance under a favorable simulation setting and to one practically useful but simplified empirical workflow. Future work should examine higher-dimensional systems, heterogeneity in innovation processes, missing data and irregular spacing, model misspecification, and Stage 1 models beyond observed-variable VARs, including latent-variable, state-space, continuous-time, and nonlinear formulations. It will also be important to study the behavior of MetaVAR when Stage 1 sampling covariance matrices are themselves approximate, such as when they are obtained via Taylor-series approximations, sandwich estimators, or bootstrap procedures.

The heterogeneity index I^2 also has limits in intensive longitudinal applications. It expresses estimated between-person variance relative to the typical Stage 1 sampling variance (Equation 8), so it describes relative precision rather than how much a dynamic feature varies across people (Borenstein et al.,

2017; Higgins & Thompson, 2002; Rücker et al., 2008). Long individual series make person-specific estimates precise, and I^2 then approaches one whether between-person variation is substantively large or modest. In the ADID illustration, every I^2 exceeded .93, so the index carried little information that the between-person standard deviations did not. For this reason, we recommend reporting the elements of $\boldsymbol{\tau}^2$ on the scale of each feature, as standard deviations with interval estimates, together with prediction intervals that describe the range of a new person's parameters (Borenstein, 2024; Higgins et al., 2009; IntHout et al., 2016). Multivariate generalizations of I^2 summarize relative heterogeneity across a parameter system (Jackson et al., 2012), but they share the same dependence on Stage 1 precision. One possible extension is a heterogeneity index scaled to the dynamic system itself, for example, between-person variation in inertia relative to the range of stable autoregressive values, or variation in implied stability and impulse-response summaries. Such indices would allow heterogeneity to be compared across studies that differ in series length and sampling frequency (Schneider & Junghaenel, 2023).

Dimensionality imposes both statistical and computational limits. With k variables, synthesizing set points and transitions gives $p = k(k + 1)$ features; including the unique innovation-covariance elements gives $p = 3k(k + 1)/2$. An unstructured between-person covariance requires $p(p + 1)/2$ components in addition to p means. For the full feature vector, this increases from 45 covariance components at $k = 2$ to 1,035 at $k = 5$. The simulation instead fitted separate set-point and transition blocks with 13 covariance components, whereas the empirical analysis used the full nine-feature covariance. With more variables, limited series length may compromise Stage 1 estimation, and limited numbers of participants may compromise estimation of the Stage 2 covariance, increasing imprecision or boundary problems. Parallel Stage 1 computation does not resolve these identification and information constraints. Diagonal or block restrictions and theory-selected covariances can reduce the burden, but impose assumptions about which features vary together. Sparse or low-rank alternatives are directions for further development. The bivariate evaluation does not demonstrate performance in higher-dimensional systems.

Several extensions are especially promising. Because Stage 2 is formulated as an SEM on the parameter system itself, person-level covariates can be used to explain why some individuals show stronger inertia, weaker cross-regulation, or different innovation structures than others, and distal outcomes can be regressed on the dynamic signature under full propagation of Stage 1 uncertainty. In the affect-dynamics context, a natural next step is to examine whether person-level emotion-regulation markers or related individual differences account for heterogeneity in set points, inertia, coupling, and innovation parameters, and whether those dynamic signatures predict clinically or developmentally meaningful endpoints. More broadly, the same framework can be used to synthesize derived functionals rather than raw parameters,

such as stability summaries, coupling contrasts, impulse-response quantities, or other theory-relevant summaries of a dynamic system.

Overall, MetaVAR contributes a computationally efficient and principled framework for drawing population-level conclusions from person-specific dynamics. By carrying the full Stage 1 sampling covariance into a multivariate population model, it separates uncertainty in idiographic estimates from variation and co-variation across people. The simulations show the inferential value of this distinction for transition heterogeneity, and the benchmark and empirical illustration demonstrate its practical appeal alongside joint Bayesian modeling. With adequate individual information and attention to variance boundaries and covariance complexity, MetaVAR enables researchers to study not only typical dynamics, but also how people differ and how those differences cohere across a dynamic system.

Open Practices Statement

This study was not preregistered. To support transparency, reproducibility, and reuse, the research compendium and analysis materials are publicly available on the Open Science Framework (https://osf.io/uenr7/overview?view_only=66801f74439442ba89a27289d7f67fab) and GitHub (https://github.com/jeksterslab/manMetaVAR; https://jeksterslab.github.io/manMetaVAR/index.html). The research compendium associated with this manuscript corresponds to `manMetaVAR` version `0.5.8`. Source code and documentation for the `fitVARMxID` and `metaDyn` R packages are also publicly available on GitHub (https://github.com/jeksterslab/fitVARMxID; https://jeksterslab.github.io/fitVARMxID/index.html; https://github.com/jeksterslab/metaDyn; https://jeksterslab.github.io/metaDyn/index.html). In addition, to facilitate reproducibility, we provide containerized computing environments with the required software dependencies. Detailed instructions for obtaining and using these containers are available at https://jeksterslab.github.io/manMetaVAR/articles/containers.html.

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Table 1

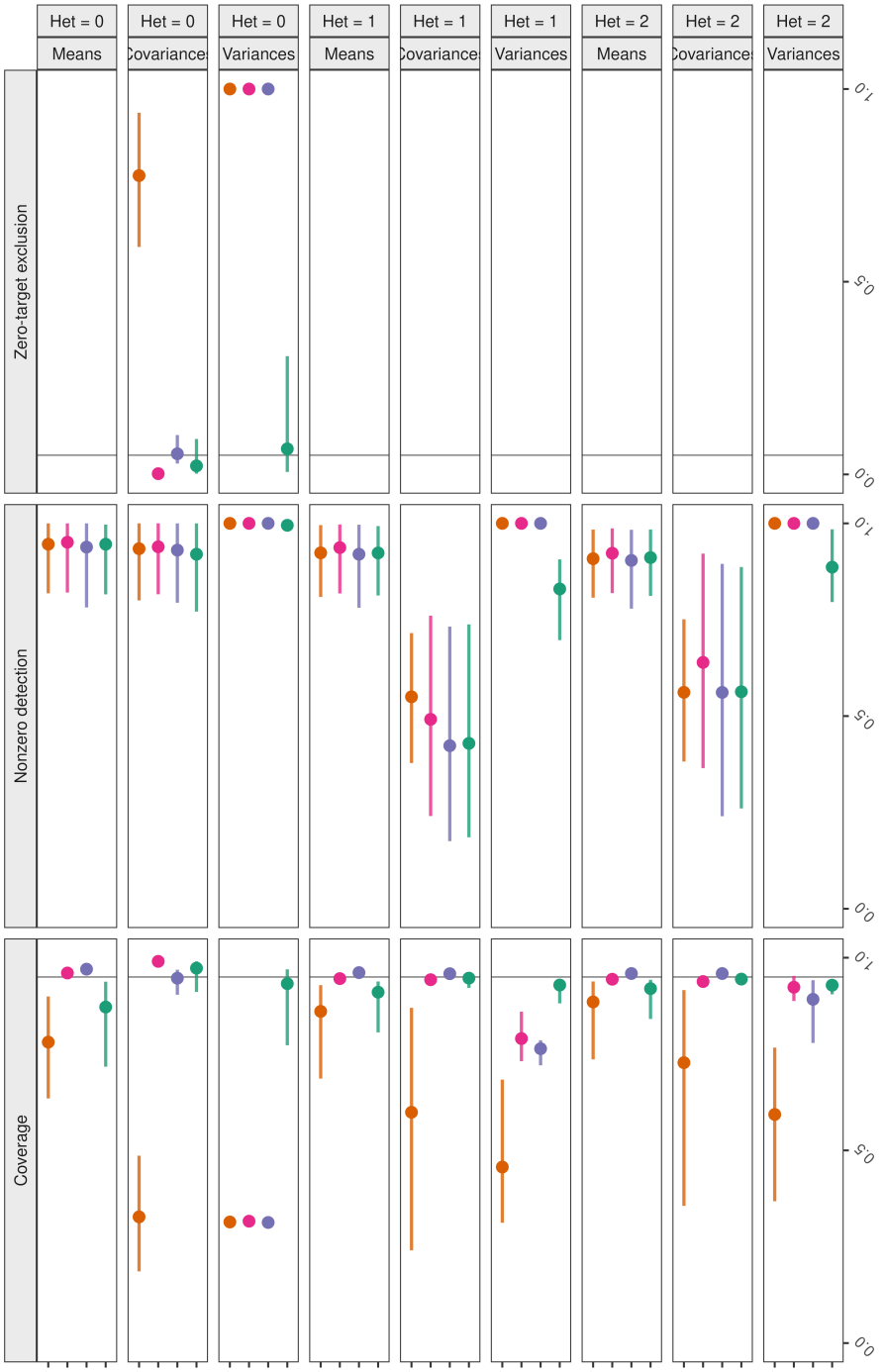
Population Means and Absolute Heterogeneity in the Empirical Illustration

Feature	MetaVAR mean [95% CI]	Uncorr mean [95% CI]	Meta SD	Uncorr SD	I^2
μ_{NA}	1.442 [1.392, 1.491]	1.443 [1.393, 1.492]	0.366	0.368	0.9989
μ_{PA}	2.573 [2.510, 2.636]	2.573 [2.510, 2.636]	0.467	0.469	0.9973
a_{11}	0.383 [0.321, 0.444]	0.343 [0.284, 0.402]	0.436	0.437	0.9662
a_{21}	-0.095 [-0.161, -0.029]	-0.097 [-0.173, -0.020]	0.418	0.567	0.9329
a_{12}	-0.039 [-0.075, -0.002]	-0.047 [-0.083, -0.011]	0.241	0.270	0.9483
a_{22}	0.329 [0.266, 0.391]	0.286 [0.228, 0.343]	0.435	0.429	0.9514
ξ_{11}	0.032 [0.028, 0.035]	0.039 [0.034, 0.044]	0.025	0.039	0.9999
ξ_{21}	-0.008 [-0.010, -0.006]	-0.011 [-0.015, -0.007]	0.011	0.027	0.9992
ξ_{22}	0.062 [0.057, 0.068]	0.070 [0.063, 0.078]	0.039	0.056	0.9961

Note. Both methods use the same nine-feature Stage 1 estimates. Meta SD is the estimated between-person standard deviation; Uncorr SD describes total dispersion without separating Stage 1 sampling uncertainty. I^2 is MetaVAR's precision-relative index. CIs are normal-theory intervals. DSEM estimates under default, alternative, and two data-informed priors are reported in the supplement.

Figure 1

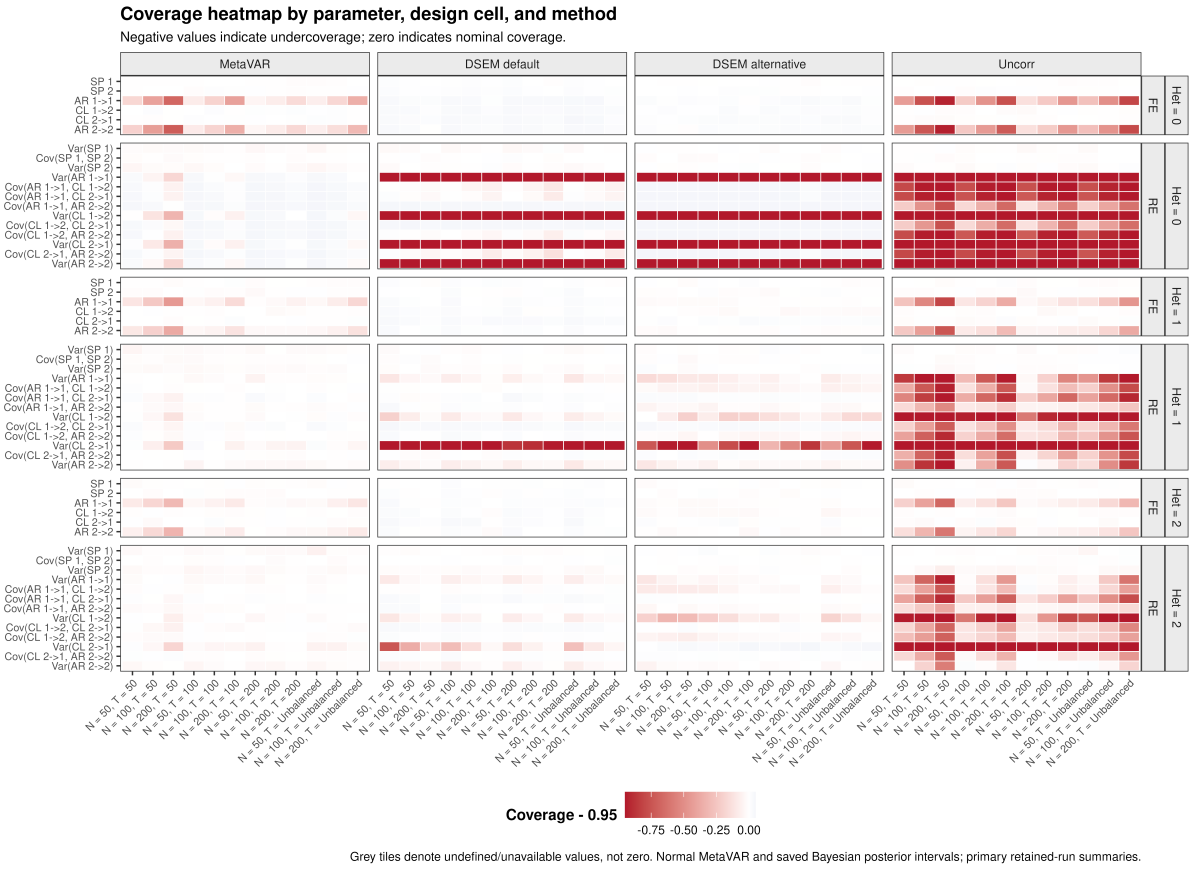
Interval Performance by Target, Heterogeneity Condition, and Method



Note. Panels separate population means, covariances, and variances, with transition-heterogeneity multipliers 0, 1, and 2. Points average parameters within each cell and then average equally across cells; ranges span cell means and are not Monte Carlo confidence intervals. Coverage uses nominal 95% intervals; detection concerns nonzero targets and zero exclusion concerns exact-zero targets. Positive-support DSEM variance intervals do not test a zero variance. MetaVAR uses normal-theory intervals; both DSEM prior branches are shown. Results are conditional on usable estimates, without posterior-quality filtering.

Figure 2

Coverage Across Transition-Heterogeneity Conditions



Note. Coverage of calibrated targets by nominal 95% intervals at transition-heterogeneity multipliers 0, 1, and 2, separately for each parameter, method, and sample-size/series-length pattern. MetaVAR uses normal-theory intervals. Both DSEM prior branches use credible intervals. Uncorr omits Stage 1 sampling uncertainty. Unequal lengths cycle through 50, 100, and 200. Companion displays for bias, RMSE, detection, and zero-target exclusion appear in the supplement.